D07 · Cross-Docking & Flow-Through

03 · Flow-through for retail replenishment

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1 WHAT IT IS

Flow-through is a retail replenishment model in which supplier volume arrives at the DC already allocated to stores, is sorted and shipped the same day, so the distribution node holds almost no inventory.

2 WHY IT MATTERS

For high-rotation categories replenished daily, the node was mostly handling, not buffering, so flow-through removes cost. **The variability that DC inventory absorbed does not disappear; it moves to the store shelf and to the supplier.** Applied to slow movers, it removes risk pooling and pushes buffer stock into every store at once.

3 HOW IT IS DONE

1. **Choose categories by rotation.** High-rotation, daily-replenished items flow through; the long tail stays on storage.
2. **Allocate upstream.** Store quantities decided before the supplier ships.
3. **Size sorting for the peak.** There is no overnight buffer; average-based sizing fails on busy days.
4. **Watch shelf availability.** It is the real outcome metric, alongside store delivery window compliance.

IN PRACTICE

Apply flow-through to rotation, not to the whole catalog

The node earns its cost by pooling variability for slow movers. For fast movers it is mostly handling. Flow-through should remove the handling and leave the pooling in place.

Size the sorting capability against the daily peak

Conventional replenishment lets you catch up tomorrow. Flow-through does not — what is not sorted today misses the store window. Average-based sizing guarantees periodic failure.

Track store on-shelf availability as the real outcome metric

Node DIO reduction is easy to celebrate and easy to achieve by pushing the problem to the shelf. If shelf availability degrades, the inventory was not eliminated — it was relocated to somewhere it works less well.

Flow-through makes supplier reliability a store-level issue

With no node buffer, a late supplier is no longer an inventory problem absorbed by the DC — it is an empty shelf. Supplier lead-time reliability must be qualified before a category moves to flow-through, not diagnosed afterward.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Flow-through for convenience store replenishment** — A distribution node serving a high-density convenience network. High-rotation categories with daily store replenishment.

Design decision	Choice	Consequence
Where store allocation is decided	Upstream, before the supplier ships	DC sorts rather than picks · Put-away and picking removed from the flow
Categories on flow-through	High-rotation, daily-replenished	Node DIO near zero · Variability absorbed by shelf and supplier

Design decision	Choice	Consequence
Categories kept on conventional replenishment	Slow-moving and long-tail	Node keeps pooling variability where the store cannot
Sorting capacity sizing	Sized to daily peak, not average	Nothing buffers overnight — a peak that exceeds sort capacity misses the store window

Result (illustrative): The model works precisely where rotation is high enough that the node was never really buffering — only handling. The failure mode is applying it to the long tail, where the node was doing real risk pooling and removing it pushes buffer stock onto every store at once.

HOW IT IS MEASURED

Flow-Through Rate % (store volume shipped without node storage)

Flow-Through Rate %(Store-destined volume dispatched the same day it was received / Total store-destined volume) × 100

Benchmark: Determined by category rotation and supplier reliability · Measure against the volume you designed for flow-through, not total volume Read this metric alongside store on-shelf availability. A rising flow-through rate with falling shelf availability means the node stopped buffering variability that the store cannot absorb.

Store Delivery Window Compliance %

Store Delivery Window Compliance %(Store deliveries arriving inside the agreed receiving window / Total store deliveries) × 100

Benchmark: High — with no overnight buffer, a missed window becomes a next-day shelf gap In flow-through there is no inventory standing by to cover a miss. Window compliance is not a service nicety; it is the mechanism that keeps the shelf full.

WHAT YOU WOULD USE

Retail Flow-Through

SAP EWM — Cross-Docking + Store Allocation — Module: Flow-Through with Upstream AllocationSAP EWM cross-dock and allocation flows for store-destined volume allocated before receipt, sorted at the node and dispatched the same day.

Blue Yonder / Manhattan Associates — Retail Replenishment — Module: Store Replenishment + Flow-Through AllocationRetail replenishment platforms that compute store-level allocation upstream and drive the sort at the distribution node.

4 SO WHAT

Apply flow-through by rotation, not to the whole catalog. Allocate upstream, size sorting for the peak and judge success by store shelf availability, not node inventory.

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